Université de Genève

Départment d'informatique

Éléments de la Théorie de l'information

Information Theory for Data Science and Machine Learning 04.03.2020

Problem Set 2 Due: 11.03.2020

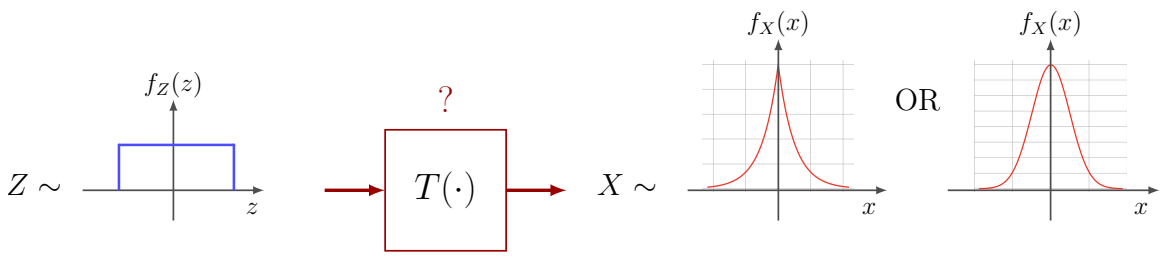
Note: Please follow the instruction on `http://sip.unige.ch/education/information-theory/`.

Problem 1 (**Transformation of Random Variables—40 points**).

a) Derive the analytical transformation $X = T(Z)$ of 1-dimensional uniform Random Variable (RV) $Z \sim f_Z(z) = \mathcal{U}(-a, a)$ into Gaussian RV $X \sim f_X(x) = \mathcal{N}\left(0, \sigma^2\right)$ or Laplacian RV $X \sim f_X(x) = \mathcal{L}(0, b)$, where:

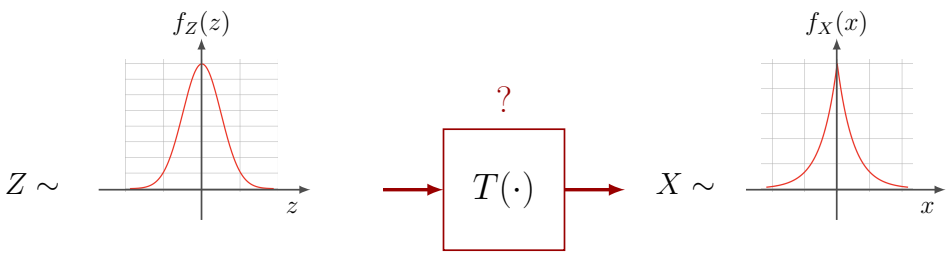
$$\begin{aligned}
\mathcal{U}(a,b) &= \begin{cases} \frac{1}{b-a}, & \forall x \in [a,b] \\ 0, & \text{otherwise} \end{cases}, \\
\mathcal{N}\left(m, \sigma^2\right) &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right), \forall x \in \mathbf{R}, \\
\mathcal{L}(m,b) &= \frac{1}{2b} \exp\left(-\frac{|x-m|}{b}\right), \forall x \in \mathbf{R}.
\end{aligned}$$

The figure below explains the expected transformation.



Provide $T(\cdot)$ for each case.

b) Drive the analytical transform, to transform Gaussian into Laplacian pdf.



What is the transformation $X = T(Z)$?

Problem 2 (**Effect of Randomization—20 points**).

In this problem, our goal is to investigate the effect of adding independent noise $\mathbf{Z}$ to a correlated signal $\mathbf{X}$. To this end, generate random vector $\mathbf{X} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}\right)$, then add the independent random noise $\mathbf{Z} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{\mathbf{z}}^2 & 0 \\ 0 & \sigma_{\mathbf{z}}^2 \end{bmatrix}\right)$ to $\mathbf{X}$. Now, investigate the effect of $\sigma_{\mathbf{z}}^2$ on the 2-dimensional pdf of random vector $\mathbf{Y} = \mathbf{X} + \mathbf{Z}$. Investigate your results for $\sigma_{\mathbf{z}}^2 = 0.1, 0.5, 1, 2, 10$ by plotting 1000 realizations. i.e., randomly generated vectors, for $\mathbf{X}$ and $\mathbf{Y}$. Describe your observations and conclusions. Indicate $\sigma\mathbf{z}^2$ after which we do not observe any more a correlation properties in $\mathbf{Y}$.

Problem 3 (**Markov's Inequality and Chebyshev's Inequality—40 points**).

a) (Markov's inequality) For any nonnegative random variable X and any $t > 0$, show that

$$\Pr\{X > t\} \leq \frac{\mathbb{E}[X]}{t}$$

Give an example of a random variable that achieves this inequality with equality.

b) (Chebyshev's inequality) Let Y be a random variable with mean m and variance σ^2. By letting $X = (Y - m)^2$, show that for any $\epsilon > 0$,

$$\Pr\{|Y - m| \geq \epsilon\} \leq \frac{\sigma^2}{\epsilon^2}.$$

c) (Weak law of large numbers) Let $Z_1, Z_2, ..., Z_n$ be a sequence of i.i.d. random variables with mean m and variance σ^2. Let $\bar{Z}_n = \frac{1}{n}\sum_{i=1}^{n} Z_i$ be the sample mean. Show that:

$$\Pr\{|\bar{Z}_n - m| \geq \epsilon\} \leq \frac{\sigma^2}{n\epsilon^2}.$$

Thus, $\Pr\{|\bar{Z}_n - m| \geq \epsilon\} \to 0$ as $n \to \infty$. This is known as the weak law of large numbers.